

Itinderjot Singh

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A. Education

Ph.D.	Atmospheric Sciences	The University of Illinois, Urbana, IL, USA	Jan 2022
M.S.	Atmospheric Sciences	The University of Illinois, Urbana, IL, USA	Dec 2016
M.Tech.	Environmental Eng.	National Institute of Technology, Kurukshetra, India	May 2014
B.E.	Civil Engineering	Birla Institute of Technology & Science Pilani, India	Dec 2010

Other Certifications:

- MITx 6.86x: Machine Learning with Python - From Linear Models to Deep Learning May-Sept 2024

B. Research and Teaching Experience

B.1 Postdoctoral Fellow – Colorado State University, Ft. Collins, CO

- Current Project: NASA INCUS Satellite Mission Jan 2023–Present
(<https://incus.colostate.edu/>)
Supervisor: Prof. Susan van den Heever
- Past Project: *Investigating Upscale Effects of Multi-day Convection in Argentina on Synoptic-scale Flow using Potential Vorticity Diagnostics* Feb 2022–Jan 2023
Supervisor: Prof. Russ Schumacher

B.2 Graduate Assistant/Visiting Student Researcher

- Graduate Research and Teaching Assistant – University of Illinois, Urbana IL Aug 2015–Jan 2022
 - Teaching Assistant for courses: (1) *Atmospheric Thermodynamics*
(2) *Computing and Data Analysis*
- ASP Graduate Visitor – MMM laboratory, NCAR, Boulder, CO May 2019–Aug 2019
 - Project: *Severe convection in Southeastern South America; RELAMPAGO (NSF-funded)*; Host: Dr. Chris Davis
- Graduate Visitor – MMM laboratory, NCAR, Boulder, CO May 2018–Aug 2018
 - Project: *Severe convection in Southeastern South America; RELAMPAGO (NSF-funded)*; Host: Dr. Chris Davis
- Graduate Research and Teaching Assistant – University of Arizona, Tucson AZ Aug 2014–Jul 2015
 - Teaching Assistant for course: *Introduction to Weather and Climate*
- Graduate Research Assistant – National Institute of Technology, Kurukshetra, India Aug 2013–May 2014
 - Project: *Analysis of Land-Use Change Dynamics and Implications of Urbanization for Environment in Chandigarh Area using Remotely Sensed Data*

C. Field work

- US NSF-funded ‘RELAMPAGO-CACTI’ project in Córdoba, Argentina Nov–Dec 2018
https://www.eol.ucar.edu/field_projects/relampago
- NASA-funded ‘TIME-SLICE’, Colorado, USA Apr–Jul 2024
<https://incus.colostate.edu/time-slice>

D. Publications

D.1 Peer-Reviewed Journal Articles

Singh, I., Bukowski, J., Marinescu, P., van den Heever, S. (2026). Environmental Controls on the Dependence of Convective Mass Flux on Model Grid Spacing. *Atmospheric Science Letters*. (In Preparation)

Singh, I., Bukowski, J., Marinescu, P., van den Heever, S. (2026). How Does Model Grid Spacing Affect the Horizontal Spatial Structure of Atmospheric Fields? *Journal of Advances in Modeling Earth Systems*. (Submitted Oct. 2026)

Singh, I., Bukowski, J., Marinescu, P., Grant, L., van den Heever, S. (2025). How Spatially Variable are Tropical and Subtropical Convective Environments? *Geophysical Research Letters*, 52, e2025GL116613.

<https://doi.org/10.1029/2025GL116613>

Singh, I., Nesbitt, S., Davis, C. (2022). Quasi-idealized numerical simulations of processes involved in orogenic convection initiation over the Sierras de Córdoba mountains. *Journal of Atmospheric Sciences*. 79, 1127–1149. <https://doi.org/10.1175/JAS-D-21-0007.1>

Singh, I., Dominguez, F., Demaria, E., & Walter, J. (2018). Extreme landfalling atmospheric river events in Arizona: Possible future changes. *Journal of Geophysical Research: Atmospheres*. 123, 7076–7097. <https://doi.org/10.1029/2017JD027866>

Marinescu, P., **Singh, I.**, Leung, G., Bukowski, J., Grant, L., van den Heever, S., (2026) Vertical Velocity Budgets in Strong, Maritime Convective Updrafts and Its Horizontal Grid Spacing Sensitivity? *Journal of Advances in Modeling Earth Systems*. (Submitted Oct. 2026)

Scarsi, F.E., Battaglia, A., Marinescu, P., Bukowski, J., **Singh, I.**, van den Heever, S. (2026). A Monte Carlo Framework for Forward Modeling of Three-dimensional Multiple Scattering Effects. *Journal of Geophysical Research: Atmospheres*. (Submitted April 2026; Under Revision)

Leganes, LJ, Prasanth, S., Navarro, A., Turk, J., Lee, G., Haddad, Z., Bukowski, J., Marinescu, P., **Singh, I.**, van den Heever, S., Tapiador, Francisco J. (2025). Machine Learning-Based Forecasting of Ice Water Path for Storm Detection and Tracking Using Geostationary Infrared and Passive Microwave Data. *Earth and Space Science*. <https://doi.org/10.1029/2025EA004406>

Marinescu, P., van den Heever, S., Grant, L., Bukowski, J., **Singh, I.** (2024). Convective Environments: How Much Sub-Grid Spatial Variability is Missing within Atmospheric Reanalysis? *Geophysical Research Letters*, 51, e2024GL111856. <https://doi.org/10.1029/2024GL111856>

D.2 Select First-authored Conference Presentations

Itinderjot Singh, J. Bukowski, P. J. Marinescu, L. D. Grant, R. Schulte, G. R. Leung, and S. C. Van Den Heever. (2026). “**Environmental Controls on the Sensitivity of Convective Mass Flux to Model Grid Spacing**” In *2026 AMS Madison Summit*. Madison, WI. <https://ams.confex.com/ams/2026Summit/meetingapp.cgi/Paper/483175>

Itinderjot Singh, Jennifer Bukowski, Peter Marinescu, Leah D Grant, Rachel L Storer, Gabrielle R Leung and Sue van den Heever. (2025). “**How Does Model Grid Spacing Affect the Horizontal Spatial Variability of Atmospheric Fields?**”. In *AGU Fall Meeting*. New Orleans, LA. <https://agu25.ipostersessions.com/?s=70-ED-9E-D8-EC-12-7C-2E-9E-87-DB-CB-CF-C2-80-AA>

Peter Marinescu, Jennie Bukowski, **Itinderjot Singh**, Leah D Grant, Rachel L Storer, Gabrielle R Leung, Sai Prasanth, Rick Schulte, Brenda Dolan, Derek J Posselt, Kristen L. Rasmussen, and Susan C. van den Heever. (2025). “**High-resolution Numerical Simulations of Tropical Convection for the NASA INCUS Mission**”. In *AGU Fall Meeting*. New Orleans, LA. <https://agu25.ipostersessions.com/?s=B4-E4-EA-75-62-AA-96-08-90-B9-0A-34-11-CE-6C-82> (Abstract submitted by P. Marinescu but poster prepared and presented by I.Singh)

Singh, Itinderjot, Jennie Bukowski, Peter J Marinescu, Leah D Grant, Gabrielle Leung, and Susan C van den Heever. (2025). “**Dependence of Horizontal Spatial Variability on Model Grid Spacing**”. In *21st Conference on Mesoscale Processes*. Boise, Idaho. <https://ams.confex.com/ams/52BC21Meso/meetingapp.cgi/Paper/461101>

Itinderjot Singh, Jennie Bukowski, Peter J Marinescu, Leah D Grant, Rachel Storer, Gabrielle R Leung, and Susan C van den Heever. (2024). “**High-resolution Numerical Simulations of Tropical Convection for the NASA INCUS Mission: Preliminary Analysis**”. In *Joint WRF/MPAS Users Workshop*. Boulder, CO. <https://files.mmm.ucar.edu/events/workshops/wrf-mpas/2024/high-resolution-numerical-simulations-of-tropical-convection.pdf>

Itinderjot Singh, Peter J Marinescu, Sean Freeman, Susan C van den Heever, Derek J Posselt. (2023). “**Using *tobac* to identify updrafts and thermals in Large Eddy Simulations of Deep Convection**”. In *Cloud Tracking Workshop* organized by GEWEX Aerosol Precipitation (GAP) initiative (<https://www.gewex.org/gap/>). Oxford, U.K (Attended virtually)

Itinderjot Singh, Russ Schumacher, Stephen W Nesbitt. (2023). “**Upscale Feedbacks from an Episode of Multiple Mesoscale Convective Systems in Argentina**”. In *American Meteorological Society Annual Meeting*. Denver, CO. <https://ams.confex.com/ams/103ANNUAL/meetingapp.cgi/Paper/420682>

Singh, Itinderjot, Stephen W Nesbitt. (2022). “**Effects of Low-Level Jet Strength on Convective Evolution near the Sierras de Córdoba Mountains, Argentina**”. In *20th Conference on Mountain Meteorology Virtual Meeting*. <https://ams.confex.com/ams/20MOUNTAIN/meetingapp.cgi/Paper/402874>

Singh, Itinderjot, Stephen W Nesbitt, and Christopher A Davis. (2020). “**Processes Controlling the Location of Convection Initiation over Complex Terrain in a Sheared, Conditionally Unstable Environment**”. In *19th Conference on Mountain Meteorology Virtual Meeting*.

Singh, Itinderjot, Stephen W. Nesbitt, and Christopher A Davis. (2019). “**High-Resolution Simulations of Orographic Convection Initiation over the Sierras de Córdoba Mountains**”. In *18th Conference on Mesoscale Processes*. Savannah, GA.

Singh, Itinderjot, and Francina Dominguez. (2016). “**Extreme Landfalling Atmospheric River Events in Arizona: Possible Future Changes**”. In *AGU Fall Meeting*. San Francisco, CA.

E. Invited Talks/Guest Lectures

- “Atmospheric Rivers”: MET 100: Meteorology (Fall 2025; taught by Prof. Aryeh Drager), State University of New York (SUNY) Oswego, NY. Dec. 2025
- “Deep moist convection in South America”: Parkland Science Club, Parkland College, Champaign, IL. Oct. 2018

F. Recognitions and Awards

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| ○ Member of AMS Committee on Mesoscale Processes | Jan 2025 - present |
| ○ First Prize Poster at the AMS 19 th Conference on Mountain Meteorology | July 2020 |
| ○ Second Prize Poster at the AMS 18 th Conference on Mesoscale Processes | Jul-Aug 2019 |
| ○ University of Illinois Graduate Conference Travel Award | Aug 2019 |
| ○ First Place in Oral Presentation at MSCAR Student conference at the Uni. of Illinois | Oct 2018 |
| ○ Graduate Aptitude Test in Engineering (GATE) Fellowship by MHRD, Govt. of India | 2012–2014 |
| ○ Birla Institute of Technology & Science Pilani Merit-Need Scholarship | 2007–2010 |

G. Professional Service

- Co-chair/Organizer of the upcoming AMS 22nd Conference on Mesoscale Processes, 2027
- Judge for student presentations and posters at the 2026 AMS Madison Summit, Madison, WI, Aug. 2026
- Member of Scientific Review Committee for IEEE GRSS International Geoscience and Remote Sensing Symposium (IGARSS) 2026
- Co-convener for session “*Developments in Convective-Scale Data Assimilation, Machine Learning, and Observations*” at the General Assembly of the European Geosciences Union in Vienna, Austria, May 2026
- Co-convener for session “*The INCUS Mission: Convective Mass Flux, Updraft Dynamics, and Anvils*” at the 21st Conference on Mesoscale Processes, June 2025
- Judge for student presentations and posters at the 21st Conference on Mesoscale Processes, Boise, ID
- Co-organizer for the workshop *AI Foundation Models for Atmospheric Science* at CSU/CIRA, August 2025
- Chaired many sessions at conferences over the years

H. Community Outreach

NASA INCUS mission outreach

- Presented about the INCUS mission and general meteorological observations to high school and community college Aerospace Interns at CSU, Ft. Collins, CO (June 2026)
- Presented about the INCUS mission to state legislators and industry representatives on the Aerospace Day at the State Capitol, Denver, CO (March 2026)
- Presented about the INCUS mission to Colorado State University alumni and families at the National Western Stock Show in Denver, CO (January 2026)
- Presented on Thunderstorms/Extreme Weather at the Wings over the Rockies Air and Space Museum in Denver, CO (Jun 6, 2025)
- Presented about the INCUS mission to legislators and industry representatives on the Aerospace Day at the State Capitol, Denver, CO (March 2025)
- Presented about the INCUS mission to Colorado State University alumni and families at the National Western Stock Show in Denver, CO (January 2025)
- Presented about the INCUS mission and general Earth observation satellites to 50 high school Aerospace Interns at CSU Spur facility, Denver, CO (August 2024)
- Presented about the INCUS mission to the public at 2023 Colorado Drone Show in Fort Collins, CO

Others

- Assisted in SEA-POL radar assembly at Steamboat Springs in November 2024 for the S2noCliME field campaign
- Wrote the *Meteorology* event for Illinois Science Olympiad Regional Tournament 2019 at Oakland Community College, March 2019
- Illinois Promise Mentor 2018 – academic, professional, and personal mentoring of a first-generation freshman at the University of Illinois Urbana-Champaign
- Involved in writing, administering, and grading the *Meteorology* event at the Illinois Science Olympiad State Competition 2018
- Teaching school kids about weather and climate at elementary and middle schools in communities within and around Fort Collins, Colorado and Urbana, Illinois

I. Media Coverage

1. <https://www.atmos.colostate.edu/2025/07/incus-researchers-present-at-amss-21st-conference-on-mesoscale-processes/>
2. <https://www.atmos.colostate.edu/2025/11/incus-scientific-team-celebrates-upcoming-mission-launch/>
3. <https://engr.source.colostate.edu/findings-in-storm-modeling-research-paper-will-inform-incus-satellite-mission/>

J. Reviewer for:

- Journal of Geophysical Research: Atmospheres
- Journal of Atmospheric Sciences
- Monthly Weather Review
- Atmospheric Science Letters
- Journal of Operational Meteorology
- IEEE International Geoscience and Remote Sensing Symposium (IGARSS) 2026

K. Professional Development

- CADRE-NOAA EPIC data assimilation training and science workshop, Norman, OK June 2026
- AI+Science Summer School at the University of Chicago Aug 2022
- Science Communication and Policy Workshop at the University of Illinois Urbana-Champaign ; organized by the American Geophysical Union Sept 2018
- XSEDE workshop on Big Data at NCSA, Urbana, IL Sept 2018

L. Professional Organizations

- Member, American Meteorological Society 2015–current
- Member, American Geophysical Union 2016–current
- Member, European Geophysical Union

M. Languages

- *Native:* Punjabi
- *Full Professional Proficiency:* Hindi, English
- *Intermediate Proficiency:* French (DELF B2), Spanish (DELE A1)

N. Professional References

1. Dr. Susan van den Heever (<https://vandenheever.atmos.colostate.edu/>)
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2. Dr. Steve Nesbitt (<https://swnesbitt.com/>)
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3. Dr. Russ Schumacher (<http://schumacher.atmos.colostate.edu/>)
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4. Dr. Francina Dominguez (<https://francina.web.illinois.edu/>)
+1 (217) 265 5483
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5. Emily Wilmsen (Director, Communications and Strategic Marketing at CSU Walter Scott, Jr. College of Engineering; Reference for public outreach events)
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